

HAN ZHANG

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[han10th.github.io](https://github.com/han10th)

WORKING EXPERIENCE

Brown University, Rhode Island
Division of Applied Mathematics
Supervised by [George Karniadakis](#),

August 2026 - Ongoing
Postdoctoral Research Associate

EDUCATION

City University of Hong Kong, Hong Kong
Department of Mathematics
Supervised by [Raymond H. CHAN](#),
co-supervised by [Xue-Cheng TAI](#) and [Jean-Michel Morel](#)

July 2026
Ph.D. in Mathematics

Chinese University of Hong Kong, Hong Kong
Department of Mathematics
Supervised by [Lok Ming LUI](#)

July 2020
M.Phil. in Mathematics

Sun Yat-Sen University, Guangdong
School of Mathematics

June 2018
B.Sc. in Computational Science

RESEARCH INTEREST

Computational Fluid Mechanics: Fluid-Structure Interaction, Blood Flow Simulation
Computational Differential Geometry: Geometric Deep Learning, Deformable Model
Scientific Machine Learning: PINN method, Neural Networks
Image Science: Image Segmentation, Interactive Segmentation

VISITING SCHOLAR

Emory University : Hosted by Prof. Yuanzhe Xi.
Princeton University: Hosted by Prof. Guillermo Sapiro.

Oct. 2025
May 2025 — Oct. 2025

JOURNAL PUBLICATIONS

^Tdenotes the corresponding author.

*denotes the equal contribution.

- Learning Task-Aware Sampling with Shared Saliency through Density-Equalizing Mappings.
Tsz Lok Ip, **Han Zhang**, Lok Ming Lui^T
Submitted *Journal of Mathematical Imaging and Vision (JMIV)*
- A Bifurcation-Based Domain Decomposition Method with Neural Operators for Blood Flow Simulation .
Yuzhou Zhao, **Han Zhang**^T, J. Matias Di Martino, Jean-Michel Morel, Guillermo Sapiro
Under Revision *SIAM Multiscale Modeling and Simulation (SIMMS)*

3. Circular quasiconformal deturbulence: Geometry-based restoration from multiple turbulent frames.
 Chu Chen, **Han Zhang**, Lok Ming Lui^T
 Accepted by *Neurocomputing (NEUCOMP)*
4. Coupled Reconstruction of 2D Blood Flow and Vessel Geometry from Noisy Images via Physics-Informed Neural Networks and Quasi-Conformal Mapping.
Han Zhang^T, Xue-Cheng Tai, Jean-Michel Morel, Raymond H. Chan
 Accepted by *SIAM Journal of Imaging Science (SIIS)*.
5. Quasi-Conformal Convolution : A Learnable Convolution for Deep Learning on Simply Connected Open Surfaces.
Han Zhang, Tsz Lok Ip, Lok Ming Lui^T
 Accepted by *SIAM Journal of Imaging Science (SIIS)*, 2026.
6. Parametrized Sampling for 3D Blood Simulation in Deformable Vessels Using Physics-Informed Neural Networks.
Han Zhang, Lingfeng Li, Xue-Cheng Tai^T, Raymond H. Chan
 Accepted by *Journal of Computational and Applied Mathematics (JCAM)*, 2025.
7. Deformation-Invariant Neural Network and Its Applications on Image Classification and Restoration.
Han Zhang, Qiguang Chen, Lok Ming Lui^T
 Accepted by *Neural Network (NEU NET)*, 2025.
8. QIS : Interactive Segmentation via Quasi-Conformal Mappings.
Han Zhang, Daoping Zhang, Lok Ming Lui^T
 Accepted by *SIAM Journal of Imaging Science (SIIS)*, 2025.
9. Full 3D Blood Flow Simulation in Curved Deformable Vessels Using Conditional Physics-Informed Neural Networks.
Han Zhang, Xue-Cheng Tai^T
 Accepted by *Acta Mathematica Universitatis Comenianae (AMUC)*, 2024.
10. A Meshless Solver for Blood Flow Simulations in Elastic Vessels Using Physics-Informed Neural Network.
Han Zhang, Raymond H. Chan, Xue-Cheng Tai^T
 Accepted by *SIAM Journal of Scientific Computing (SISC)*, 2024.
11. A Learning-based Framework for Topology-Preserving Segmentation using Quasiconformal Mappings.
Han Zhang, Lok Ming Lui^T
 Accepted by *Neurocomputing (NEUCOMP)*, 2024.
12. A New Framework for Deformation Analysis with Uncertainties using Beltrami Fourier Descriptor.
Han Zhang, Lok Ming Lui^T.

Accepted by *Journal of Mathematics, Computation and Geometry of Data (MCGD)*, 2022.

13. Quasi-Conformal Neural Network (QC-net) with Applications to Shape Matching.

Han Zhang, Lok Ming Lui^T.

Accepted by *Journal of Mathematics, Computation and Geometry of Data (MCGD)*, 2021.

14. Continuous Aerial Path Planning for 3D Urban Scene Reconstruction.

Han Zhang, Yucong Yao, Ke Xie, Chi-Wing Fu, Hao Zhang, Hui Huang^T.

Accepted by *ACM Transaction on Computer Graphics (ACM TOG, SIGGRAPH ASIA)*, 2021.

PROCEEDING PUBLICATIONS

1. Fast Physics-Informed Learning via Diffusion Hypernetworks.

Yuzhou Zhao, **Han Zhang**^T, J. Matias Di Martino, Jean-Michel Morel, Guillermo Sapiro
Submitted

[AI4PDE project]

2. Nondeterministic Deformation analysis using Quasiconformal Geometry.

Han Zhang, Lok Ming Lui^T

Accepted by *IEEE International Conference on Image Processing (ICIP)*, 2022.

[Geometric Image project]

ACADEMIC ACHIEVEMENTS

Outstanding Academic Performance Award, 2024

Excellent Student Scholarship of Sun Yat-Sen University, 2017

First Class

Excellent Thesis of Sun Yat-Sen University, 2018

Outstanding

China Undergraduate Mathematical Contest in Modeling, 2016

Second Prize

National High School Mathematics League, 2012

Second Prize

INVITED TALKS

1. **[1-hour Invited Talk]** Physics-Informed Neural Network for Blood Flow Simulation :
Forward and Inverse Problem

Tsinghua University - MATH+ML+X Seminar, Online

May 2026

2. **[1-hour Invited Talk]** Physics-Informed Neural Network for Blood Flow Simulation :
Forward and Inverse Problem

Brown University - CRUNCH Seminar, Online

Mar. 2026

3. **[1-hour Invited Talk]** Physics-Informed Neural Network for Blood Flow Simulation :
Forward and Inverse Problem

Emory University - CODES Seminar, Atlanta, USA

Oct. 2025

REVIEW

1. Computer Graphics Forum
2. Journal of Mathematical Imaging and Vision
3. Medical Image Analysis
4. Neural Networks
5. Neurocomputing
6. Pattern Recognition

TEACHING EXPERIENCE

* denotes courses with duty of a weekly one-hour instructional session.

City University of Hong Kong

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|---|---------------------|
| 1. MA1200 Calculus & Basic Linear Algebra I* | <i>Spring, 2025</i> |
| 2. MA1201 Calculus & Basic Linear Algebra II* | <i>Spring, 2025</i> |
| 3. MA2177 Engineering Mathematics and Statistics* | <i>Fall, 2024</i> |
| 4. MA1200 Calculus & Basic Linear Algebra I* | <i>Spring, 2024</i> |
| 5. MA1200 Calculus & Basic Linear Algebra I* | <i>Fall, 2023</i> |
| 6. GE1359/MA1502 Algebra | <i>Spring, 2023</i> |

Chinese University of Hong Kong

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|-------------------------------------|---------------------|
| 1. MATH1010 University Mathematics | <i>Spring, 2020</i> |
| 2. MATH1510B Calculus for Engineers | <i>Fall, 2019</i> |
| 3. MATH4250 Game Theory | <i>Spring, 2019</i> |
| 4. MATH1510B Calculus for Engineers | <i>Fall, 2018</i> |